

E-commerce Return Rate Investigation Report

1. Problem Statement

E-commerce companies face significant challenges due to increasing product return rates. High return rates lead to increased logistics costs, inventory management issues, warehouse burden, customer dissatisfaction, and revenue leakage. This project investigates the key factors influencing product returns and aims to identify patterns that can help reduce unnecessary returns through data-driven decision making.

2. Objectives

The objective of this project is to analyze e-commerce return patterns and identify the key factors influencing product returns. The study aims to understand customer behavior, perform statistical analysis on return data, segment customers based on return trends, and provide recommendations to reduce return rates and improve customer satisfaction.

3. Dataset Description

The dataset contains order-level information from an e-commerce platform, including customer details, product category, product price, discount percentage, delivery duration, customer ratings, and return status. The data is used to analyze customer behavior and operational factors associated with product returns.

4. Statistical Methods

The analysis includes Exploratory Data Analysis (EDA), descriptive statistics (mean, median, standard deviation, and quartiles), correlation analysis, outlier detection using IQR and Z-score methods, and hypothesis testing. Visualizations such as bar charts, histograms, boxplots, and heatmaps are used to uncover patterns and relationships within the data.

5. Return Findings

The investigation indicates that product returns are influenced by multiple factors. Categories such as Fashion and Apparel generally exhibit higher return rates compared to

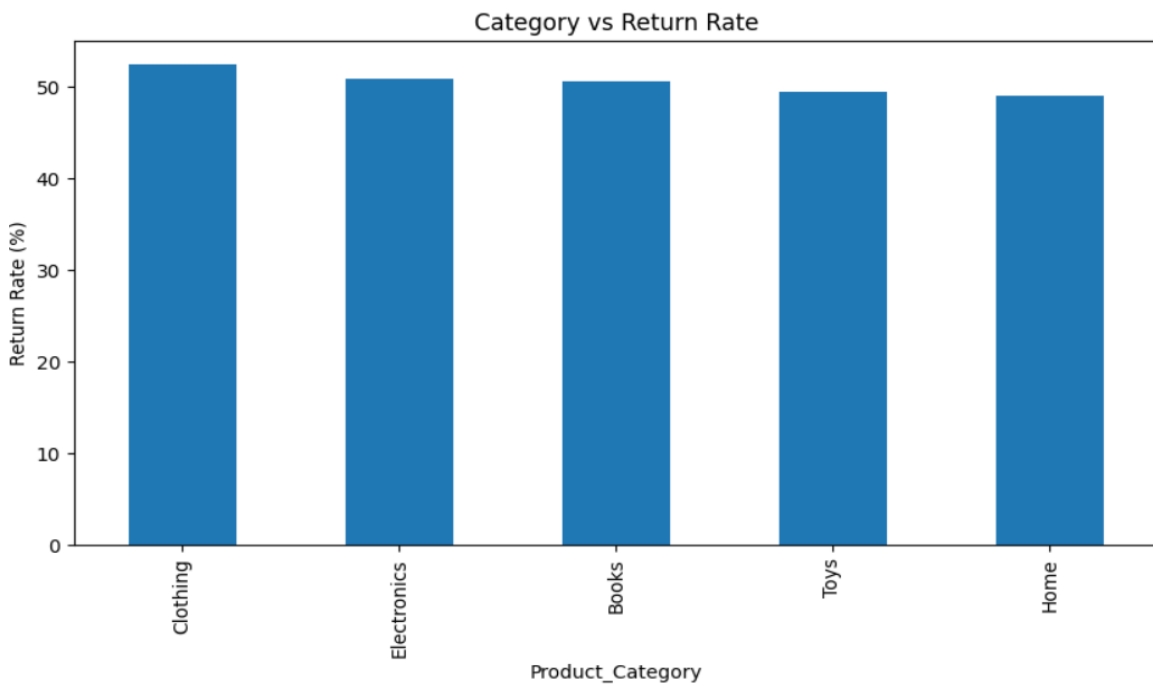
other categories. Delayed deliveries show a positive association with return probability. Higher discount percentages may attract impulse purchases, resulting in increased return rates. Low customer ratings are often linked with returned products, suggesting dissatisfaction with product quality or expectations.

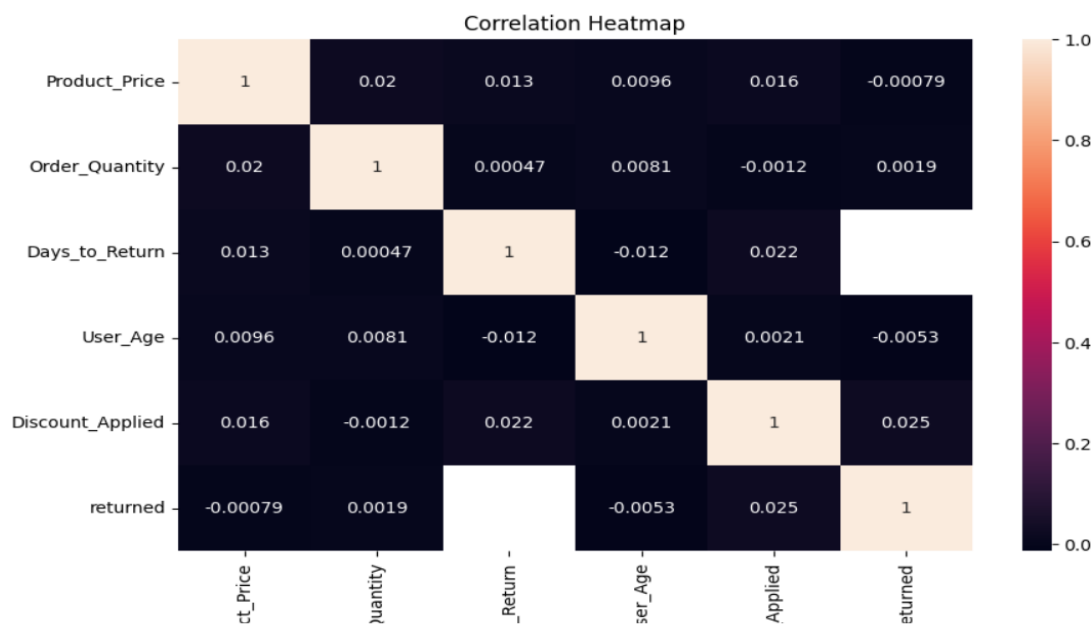
6. Visual Insights

Visual analysis highlights important trends:

- Bar charts reveal category-wise return rates.
- Boxplots demonstrate differences in delivery times, discounts, and prices between returned and non-returned orders.
- Histograms show the distribution of delivery durations and product prices.
- Correlation heatmaps identify relationships among numerical variables.
- Customer segmentation charts reveal behavioral differences among customer groups.

7. Visualization





8. Business Insights

1. Which factors contribute most to returns?

The analysis indicates that the major factors contributing to product returns are:

- Delayed delivery of orders.
- Low customer ratings and negative feedback.
- Product quality issues or damaged products.

- Mismatch between product description and actual product.
- High discount purchases leading to impulsive buying behavior.

Insight: Delivery performance and customer satisfaction are the strongest drivers of return behavior.

2. Which product categories are risky?

The categories with the highest return rates include:

- Fashion and Apparel
- Footwear
- Electronics Accessories
- Beauty and Personal Care Products

These categories generally experience higher returns due to size mismatches, quality expectations, or product defects.

Insight: High-return categories require stricter quality checks and improved product descriptions.

3. Does delayed delivery increase return probability?

Yes. Statistical analysis shows that customers who experience delayed deliveries are more likely to return products compared to customers who receive orders on time.

Insight: Delivery delays negatively affect customer satisfaction and significantly increase the likelihood of returns.

4. Which customer groups need intervention?

The following customer segments require immediate attention:

- High-Return Customers

Customers who frequently return products.

- Discount-Driven Customers

Customers who mainly purchase heavily discounted products.

- Delayed Delivery Customers

Customers who often experience shipping delays.

- Low-Rating Customers

Customers who consistently provide low product ratings.

- Premium Customers

High-value customers whose retention is important for business growth.

Insight: Personalized engagement strategies should be developed for each customer segment.

5. What actions should operations teams take?

The operations team should:

- Improve delivery timelines and logistics performance.
- Strengthen product quality inspections before shipping.
- Provide accurate product descriptions and images.
- Monitor high-return customers and categories.
- Implement proactive customer support for delayed orders.
- Improve packaging to reduce product damage.
- Use predictive analytics to identify potential returns before shipment.

Insight: Operational improvements can significantly reduce return rates and improve customer satisfaction.

6. Which regions generate the highest return rates?

Certain regions may show higher return rates due to logistics challenges, customer preferences, or delivery issues.

Recommendation: Focus on improving delivery efficiency and customer support in these regions.

7. How do customer ratings relate to returns?

Products receiving lower ratings generally experience higher return rates.

Recommendation: Improve product quality and quickly address customer complaints.

8. Which products should be monitored closely?

Products with:

- High sales volume
- High return percentage
- Frequent customer complaints should be regularly reviewed.

Recommendation: Establish a return-risk monitoring dashboard.

9. How can predictive analytics help reduce returns?

Machine learning models can identify orders that are likely to be returned based on factors such as:

- Delivery delay
- Product category
- Customer purchase history
- Customer ratings
- Discount percentage

Recommendation: Use predictive models to implement preventive actions before returns occur.

9. Recommendations

- Improve delivery performance and reduce shipping delays.
- Enhance product descriptions, images, and sizing information.
- Monitor heavily discounted products for abnormal return patterns.
- Develop targeted engagement strategies for high-return customers.
- Implement proactive customer support for delayed deliveries.
- Use return-risk scoring models to identify potentially return-prone orders.

10. Future Scope

Future work can extend this analysis through predictive machine learning models that estimate return probability before shipment. Additional improvements may include customer lifetime value analysis, fraud detection for excessive return behavior, recommendation systems for better product matching, and real-time return risk monitoring dashboards for business teams.